

	$\mathcal{K}_0^{\#1}$	$\mathcal{K}_0^{\#1}{}_{\alpha\beta\chi}$				
$\mathcal{K}_0^{\#1} \dagger$	$-\frac{3\Upsilon_1 k^2}{4}$	0				
$\mathcal{K}_0^{\#1} \dagger^{\alpha\beta\chi}$	0	$\Upsilon_2 k^2 + 3 \overset{(2)}{\kappa}_1$	$\mathcal{K}_1^{\#1}{}_{\alpha\beta}$	$\mathcal{K}_1^{\#2}{}_{\alpha\beta}$	$\mathcal{K}_1^{\#1}{}_{\alpha}$	$\mathcal{K}_1^{\#2}{}_{\alpha}$
	$\mathcal{K}_1^{\#1} \dagger^{\alpha\beta}$	$\Upsilon_3 k^2 + 2 \overset{(2)}{\kappa}_1$	$\frac{-\Upsilon_4 k^2 + 4 \overset{(2)}{\kappa}_1}{2\sqrt{2}}$	0	0	
	$\mathcal{K}_1^{\#2} \dagger^{\alpha\beta}$	$\frac{-\Upsilon_4 k^2 + 4 \overset{(2)}{\kappa}_1}{2\sqrt{2}}$	$\frac{1}{4}(-\Upsilon_5 k^2 + 4 \overset{(2)}{\kappa}_1)$	0	0	
	$\mathcal{K}_1^{\#1} \dagger^{\alpha}$	0	0	0	0	
	$\mathcal{K}_1^{\#2} \dagger^{\alpha}$	0	0	0	0	$\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta}$ $\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta\chi}$
			$\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta}$	0	0	
			$\mathcal{K}_{2^+}^{\#2} \dagger^{\alpha\beta\chi}$	0	$\frac{\Upsilon_6 k^2}{2}$	

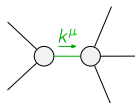
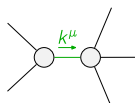
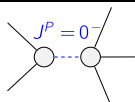
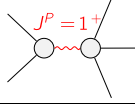
	$\mathcal{T}_{0^+}^{\#1}$	$\mathcal{T}_{0^+ \alpha\beta\chi}^{\#1}$				
$\mathcal{T}_{0^+}^{\#1} \dagger$	$\frac{1}{\text{Det}(0^+)}$	0				
$\mathcal{T}_{0^+}^{\#1} \dagger^{\alpha\beta\chi}$	0	$\frac{1}{\text{Det}(0^+)}$	$\mathcal{T}_{1^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{T}_{1^+}^{\#2}{}_{\alpha\beta}$	$\mathcal{T}_{1^+}^{\#1}{}_{\alpha}$	$\mathcal{T}_{1^+}^{\#2}{}_{\alpha}$
	$\mathcal{T}_{1^+}^{\#1} \dagger^{\alpha\beta}$	$\frac{-Y_5 k^2 + 4 \frac{(2)}{K_1}}{4 \text{Det}(1^+)}$	$\frac{Y_4 k^2 - 4 \frac{(2)}{K_1}}{2 \sqrt{2} \text{Det}(1^+)}$	0	0	
	$\mathcal{T}_{1^+}^{\#2} \dagger^{\alpha\beta}$	$\frac{Y_4 k^2 - 4 \frac{(2)}{K_1}}{2 \sqrt{2} \text{Det}(1^+)}$	$\frac{Y_3 k^2 + 2 \frac{(2)}{K_1}}{\text{Det}(1^+)}$	0	0	
	$\mathcal{T}_{1^+}^{\#1} \dagger^{\alpha}$	0	0	0	0	
	$\mathcal{T}_{1^+}^{\#2} \dagger^{\alpha}$	0	0	0	0	
				$\mathcal{T}_{2^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{T}_{2^+}^{\#1}{}_{\alpha\beta\chi}$	
			$\mathcal{T}_{2^+}^{\#1} \dagger^{\alpha\beta}$	0	0	
			$\mathcal{T}_{2^+}^{\#1} \dagger^{\alpha\beta\chi}$	0	$\frac{1}{\text{Det}(2^+)}$	

Abbreviations used in matrices

$$\begin{aligned}
Y_1 &= 2 \binom{(4)}{\kappa_1} + \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \&\& Y_2 = \binom{(4)}{\kappa_{15}} - \binom{(4)}{\kappa_{16}} \&\& Y_3 = 2 \binom{(4)}{\kappa_{15}} - \binom{(4)}{\kappa_3} \&\& Y_4 = 2 \binom{(4)}{\kappa_{16}} - \binom{(4)}{\kappa_4} \&\& \\
Y_5 &= 2 \binom{(4)}{\kappa_{15}} + 3 \binom{(4)}{\kappa_{16}} - 2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4} \&\& Y_6 = 2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \&\& \text{Det}(0^+) = -\frac{3}{4} k^2 (2 \binom{(4)}{\kappa_1} + \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}}) \&\& \\
\text{Det}(1^+) &= \frac{1}{8} (-k^4 (8 \binom{(4)}{\kappa_{15}}^2 + 4 \binom{(4)}{\kappa_{16}}^2 + 4 \binom{(4)}{\kappa_{15}} (3 \binom{(4)}{\kappa_{16}} - 3 \binom{(4)}{\kappa_3} - 2 \binom{(4)}{\kappa_4}) + (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^2 - 2 \binom{(4)}{\kappa_{16}} (3 \binom{(4)}{\kappa_3} + 2 \binom{(4)}{\kappa_4})) + \\
&4 k^2 (2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}}) \binom{(2)}{\kappa_1}) \&\& \text{Det}(2^-) = \frac{1}{2} k^2 (2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}})
\end{aligned}$$

Lagrangian

$$\begin{aligned}
& \kappa_1^{(2)} \mathcal{K}_{\alpha\beta\chi} \mathcal{K}^{\alpha\beta\chi} - 2 \kappa_1^{(2)} \mathcal{K}^{\alpha\beta\chi} \mathcal{K}_{\beta\alpha\chi} + \kappa_1^{(4)} \partial_\beta \mathcal{K}^\delta_{\chi\delta} \partial^\chi \mathcal{K}^\alpha_{\alpha\beta} + \\
& \kappa_1^{(4)} \kappa_{15} \partial_\chi \mathcal{K}^\delta_{\beta\delta} \partial^\chi \mathcal{K}^\alpha_{\alpha\beta} + \frac{1}{2} \kappa_1^{(4)} \kappa_{16} \partial_\chi \mathcal{K}^\delta_{\beta\delta} \partial^\chi \mathcal{K}^\alpha_{\alpha\beta} + \kappa_3^{(4)} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\alpha\chi}^\delta + \kappa_4^{(4)} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta + \\
& 2 \kappa_1^{(4)} \kappa_{15} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}^\delta + \kappa_1^{(4)} \kappa_{16} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}^\delta - \kappa_3^{(4)} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}^\delta - 2 \kappa_1^{(4)} \kappa_{15} \partial^\chi \mathcal{K}^\alpha_{\alpha\beta} \partial_\delta \mathcal{K}_{\chi\beta}^\delta - \\
& \kappa_1^{(4)} \kappa_{16} \partial^\chi \mathcal{K}^\alpha_{\alpha\beta} \partial_\delta \mathcal{K}_{\chi\beta}^\delta - \frac{3}{2} \kappa_1^{(4)} \kappa_{15} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} - \frac{3}{4} \kappa_1^{(4)} \kappa_{16} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} + \\
& \frac{1}{2} \kappa_3^{(4)} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} + \frac{1}{2} \kappa_4^{(4)} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} + \kappa_1^{(4)} \kappa_{15} \partial_\delta \mathcal{K}_{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} + \kappa_1^{(4)} \kappa_{16} \partial_\delta \mathcal{K}_{\beta\alpha\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi}
\end{aligned}$$

Added source term(s):		$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$	
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}_1^{\#2\alpha} == 0$	3	$\partial_\chi \partial_\beta \mathcal{J}^{\beta\alpha\chi} == 0$	
$\mathcal{J}_1^{\#1\alpha} == 0$	3	$\partial_\chi \partial^\alpha \mathcal{J}^{\beta\chi}{}_\alpha + \partial_\chi \partial^\chi \mathcal{J}^{\beta\alpha}{}_\beta == \partial_\chi \partial_\beta \mathcal{J}^{\beta\alpha\chi}$	
$\mathcal{J}_{2^+}^{\#1\alpha\beta} == 0$	5	$3 \partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 3 \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2 \eta^{\alpha\beta} \partial_\epsilon \partial^\epsilon \partial_\delta \mathcal{J}^{\chi\chi}{}_\delta == 2 \partial_\delta \partial^\beta \partial^\alpha \mathcal{J}^{\chi\chi}{}_\delta + 3 (\partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\beta\alpha\chi})$	
Total # constraint(s):	11		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	1	0	$\frac{1}{(2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}})^2 \binom{(2)}{\kappa_1}} (6 \binom{(4)}{\kappa_{15}} \binom{(2)}{\kappa_1} + 3 \binom{(4)}{\kappa_{16}} \binom{(2)}{\kappa_1} + 80 \binom{(4)}{\kappa_{15}}^2 p^2 + 80 \binom{(4)}{\kappa_{15}} \binom{(4)}{\kappa_{16}} p^2 +$ $20 \binom{(4)}{\kappa_{16}}^2 p^2 - 108 \binom{(4)}{\kappa_{15}} \binom{(4)}{\kappa_3} p^2 - 54 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_3} p^2 + 36 \binom{(4)}{\kappa_3}^2 p^2 - 54 \binom{(4)}{\kappa_{15}} \binom{(4)}{\kappa_4} p^2 - 27 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_4} p^2 +$ $36 \binom{(4)}{\kappa_3} \binom{(4)}{\kappa_4} p^2 + 9 \binom{(4)}{\kappa_4}^2 p^2 - \sqrt{(10000 \binom{(4)}{\kappa_{15}}^4 p^4 + 625 \binom{(4)}{\kappa_{16}}^4 p^4 + 81 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^4 p^4 +$ $30 \binom{(4)}{\kappa_{16}}^3 p^2 (\binom{(2)}{\kappa_1} - 45 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4}) p^2) - 54 \binom{(4)}{\kappa_{16}} (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^2 p^2 (-\binom{(2)}{\kappa_1} + 9 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4}) p^2) +$ $18 \binom{(4)}{\kappa_{16}}^2 (28 \binom{(2)}{\kappa_1} - 6 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4}) \binom{(2)}{\kappa_1} p^2 + 65 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^2 p^4) +$ $80 \binom{(4)}{\kappa_{15}}^3 (3 \binom{(2)}{\kappa_1} p^2 + 5 (50 \binom{(4)}{\kappa_{16}} - 27 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})) p^4) + 24 \binom{(4)}{\kappa_{15}}^2$ $(84 \binom{(2)}{\kappa_1}^2 + 3 (5 \binom{(4)}{\kappa_{16}} - 6 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})) \binom{(2)}{\kappa_1} p^2 + 5 (125 \binom{(4)}{\kappa_{16}}^2 - 135 \binom{(4)}{\kappa_{16}} (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4}) + 39 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^2) p^4) +$ $4 \binom{(4)}{\kappa_{15}} (1250 \binom{(4)}{\kappa_{16}}^3 p^4 + 45 \binom{(4)}{\kappa_{16}}^2 p^2 (\binom{(2)}{\kappa_1} - 45 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4}) p^2) - 27 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^2 p^2$ $(-\binom{(2)}{\kappa_1} + 9 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4}) p^2) + 18 \binom{(4)}{\kappa_{16}} (28 \binom{(2)}{\kappa_1} - 6 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4}) \binom{(2)}{\kappa_1} p^2 + 65 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^2 p^4))))$
	1	0	$\frac{1}{(2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}})^2 \binom{(2)}{\kappa_1}} (6 \binom{(4)}{\kappa_{15}} \binom{(2)}{\kappa_1} + 3 \binom{(4)}{\kappa_{16}} \binom{(2)}{\kappa_1} + 80 \binom{(4)}{\kappa_{15}}^2 p^2 + 80 \binom{(4)}{\kappa_{15}} \binom{(4)}{\kappa_{16}} p^2 +$ $20 \binom{(4)}{\kappa_{16}}^2 p^2 - 108 \binom{(4)}{\kappa_{15}} \binom{(4)}{\kappa_3} p^2 - 54 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_3} p^2 + 36 \binom{(4)}{\kappa_3}^2 p^2 - 54 \binom{(4)}{\kappa_{15}} \binom{(4)}{\kappa_4} p^2 - 27 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_4} p^2 +$ $36 \binom{(4)}{\kappa_3} \binom{(4)}{\kappa_4} p^2 + 9 \binom{(4)}{\kappa_4}^2 p^2 + \sqrt{(10000 \binom{(4)}{\kappa_{15}}^4 p^4 + 625 \binom{(4)}{\kappa_{16}}^4 p^4 + 81 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^4 p^4 +$ $30 \binom{(4)}{\kappa_{16}}^3 p^2 (\binom{(2)}{\kappa_1} - 45 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4}) p^2) - 54 \binom{(4)}{\kappa_{16}} (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^2 p^2 (-\binom{(2)}{\kappa_1} + 9 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4}) p^2) +$ $18 \binom{(4)}{\kappa_{16}}^2 (28 \binom{(2)}{\kappa_1} - 6 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4}) \binom{(2)}{\kappa_1} p^2 + 65 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^2 p^4) +$ $80 \binom{(4)}{\kappa_{15}}^3 (3 \binom{(2)}{\kappa_1} p^2 + 5 (50 \binom{(4)}{\kappa_{16}} - 27 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})) p^4) + 24 \binom{(4)}{\kappa_{15}}^2$ $(84 \binom{(2)}{\kappa_1}^2 + 3 (5 \binom{(4)}{\kappa_{16}} - 6 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})) \binom{(2)}{\kappa_1} p^2 + 5 (125 \binom{(4)}{\kappa_{16}}^2 - 135 \binom{(4)}{\kappa_{16}} (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4}) + 39 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^2) p^4) +$ $4 \binom{(4)}{\kappa_{15}} (1250 \binom{(4)}{\kappa_{16}}^3 p^4 + 45 \binom{(4)}{\kappa_{16}}^2 p^2 (\binom{(2)}{\kappa_1} - 45 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4}) p^2) - 27 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^2 p^2$ $(-\binom{(2)}{\kappa_1} + 9 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4}) p^2) + 18 \binom{(4)}{\kappa_{16}} (28 \binom{(2)}{\kappa_1} - 6 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4}) \binom{(2)}{\kappa_1} p^2 + 65 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^2 p^4))))$
	1	$\frac{3 \binom{(2)}{\kappa_1}}{-\binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}}}$	$\frac{1}{-\binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}}}$
	3	$\frac{4 (2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}}) \binom{(2)}{\kappa_1}}{8 \binom{(4)}{\kappa_{15}}^2 + 4 \binom{(4)}{\kappa_{16}}^2 + 4 \binom{(4)}{\kappa_{15}} (3 \binom{(4)}{\kappa_{16}} - 3 \binom{(4)}{\kappa_3} - 2 \binom{(4)}{\kappa_4}) + (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^2 - 2 \binom{(4)}{\kappa_{16}} (3 \binom{(4)}{\kappa_3} + 2 \binom{(4)}{\kappa_4})}$	$-\frac{2 (36 \binom{(4)}{\kappa_{15}}^2 + 9 \binom{(4)}{\kappa_{16}}^2 - 10 \binom{(4)}{\kappa_{16}} (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4}) + 3 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^2 + 4 \binom{(4)}{\kappa_{15}} (9 \binom{(4)}{\kappa_{16}} - 5 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})))}{(2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}}) (8 \binom{(4)}{\kappa_{15}}^2 + 4 \binom{(4)}{\kappa_{16}}^2 + 4 \binom{(4)}{\kappa_{15}} (3 \binom{(4)}{\kappa_{16}} - 3 \binom{(4)}{\kappa_3} - 2 \binom{(4)}{\kappa_4}) + (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^2 - 2 \binom{(4)}{\kappa_{16}} (3 \binom{(4)}{\kappa_3} + 2 \binom{(4)}{\kappa_4}))}$
Resolved unitarity condition(s):		(Demonstrably impossible)	